

STATEMENT OF OBJECTIVES (SOO)

1. Contracting Officer's Representatives (COR).

a. TBD

2. Contract Title. Mainframe and Legacy System Modernization

3. Background. DISA's mainframe and legacy computing systems are essential in providing enterprise-level IT services for the Department of Defense (DoD) and its agencies. These systems are used for many critical tasks, including managing logistics, finances, personnel, and command and control. DISA's mainframe and legacy systems host and manage critical applications, handle large amounts of data, are reliable, and meet strict DoD security requirements, making them essential for the DoD's operations.

DISA recognizes that aging IT systems create problems and modernization is required to keep critical operations running smoothly. It is becoming more expensive to maintain these legacy systems and fewer people have the technical skills to operate and maintain them. There are also limitations in using modern tools like cloud computing, AI, and advanced analytics with these systems. This limits how well they can scale, innovate, and meet the latest federal cybersecurity and data standards. Modernizing these systems will improve efficiency, allow for secure data sharing, and align with the DoD's strategic goals. By using flexible and secure modernization plans, DISA will ensure mainframes continue to support defense communications and operations.

To address these challenges, the Mainframe Line of Business (MLOB) is streamlining operations, reducing costs, and aligning mainframe and legacy systems with the DoD's strategic goals. The MLOB is responsible for maintaining legacy systems while also exploring new technologies and ways to modernize them. By connecting legacy systems with modern IT practices, MLOB strengthens DISA's mission to provide superior services, ensuring the DoD maintains its technological edge in defense communications and operations.

4. Objectives. This contract will support the modernization of a wide variety of mainframe and legacy systems. Contractor support will focus on these six objectives :

Objective 1: Enhance Operational Efficiency and Agility

Modernization efforts will make systems more scalable, efficient, and adaptable to emerging technologies. This includes moving legacy systems to cloud or hybrid cloud environments, using resources better, and enabling real-time data access for mission-critical operations.

Objective 2: Ensure Security and Compliance

Modernization efforts will improve security postures and ensure compliance with federal and DoD regulations, such as the Risk Management Framework (RMF). This involves adding robust security measures, encryption, access controls, and monitoring to protect sensitive data and operations.

Objective 3: Leverage Advanced Technologies

Modernization efforts will add advanced technologies, such as artificial intelligence (AI), automation, and API enablement to improve system capabilities and efficiency. This includes enabling real-time analytics, automating tasks, and improving decision-making with AI tools.

Objective 4: Data Modernization

Modernization efforts will transition DISA's data architecture to a more flexible, scalable, and secure model. This includes creating data lakes, implementing DataOps practices, and improving data governance to ensure data integrity, accessibility, and interoperability for mission-critical operations.

Objective 5: Address Workforce Challenges

Modernization efforts will train the workforce on how to manage new technologies and systems. This will ensure continued expertise, reduce reliance on legacy skills, and prepare the workforce to use technologies like cloud, automation, and AI, supporting long-term system sustainability.

Objective 6: Achieve Cost-Effective Modernization

Modernization efforts will optimize costs by reducing reliance on legacy hardware, streamlining infrastructure, and using pay-as-you-go cloud models. This aims to achieve a return on investment (ROI) while maintaining mission-critical functions and ensuring seamless operations during and after the transition.

5. Scope

This SOO describes the work required to modernize DISA and its Strategic Partners' legacy systems. The effort focuses on three key areas: Program Management Office (PMO) support, assessments, and implementation. Each area is critical to achieving the modernization goals and aligning with mission objectives.

Program Management Office (PMO) Support: The contractor will establish and manage PMO Support to oversee all modernization efforts. This includes providing governance, risk management, and tracking performance to meet project goals and timelines. The contractor will coordinate activities among stakeholders, including Strategic Partners and other contractors, to ensure effective communication and collaboration.

Assessments: The contractor will assess existing legacy systems to identify inefficiencies, risks, and opportunities for modernization. This includes evaluating system architecture, applications, data, and infrastructure to determine readiness for modernization. Based on these assessments, the contractor will deliver recommendations and actionable, phased roadmaps aligned with mission-critical objectives. The contractor will work with solution architects to design scalable, secure, and compliant solutions that address the unique needs of DISA and its Strategic Partners. These efforts will support informed decision-making, optimize system performance, and ensure compliance with applicable federal standards.

Implementation: The contractor will execute modernization strategies to transition legacy systems to modern, secure, and scalable solutions. This includes activities such as code refactoring, data migration, system integration, and the deployment of advanced technologies like cloud computing, artificial intelligence, and automation tools. The contractor will ensure interoperability, optimize performance, and maintain compliance with applicable federal standards. Implementation will involve phased transitions to minimize disruption, expert oversight to streamline processes, and technical support to address post-implementation challenges.

6. Tasks

The contractor will propose the following tasks to be included in the performance work statement. All deliverables will have corresponding Contract Data Requirement List (CDRL) and an appropriate Data Item Description (DID), and will be mapped to the specific task by task number to the Contractor Work Breakdown Structure (WBS) (e.g., task 6.1, WBS 1.1). The contractor can access and select appropriate Data Item Descriptions from the list of active DIDs in Defense Logistics Agency's ASSIST Portal at <https://assist.dla.mil/online/start/index.cfm>. The contractor will deliver and update a Data Accession List (DAL), under DID number DI-MGMT-81453C, listing and specifying all internally generated data and computer software used by the contractor (including subcontractors and vendors) to develop, test, and manage the program, and disclose all Data Rights Assertions (DFARS 252.227-7017).

6.1 - Project Management

Project Management activities include adhering to industry-standard project management methodologies, as agreed upon during the project initiation phase. The Contractor will be responsible for developing and maintaining a comprehensive Project Management Plan (PMP) that outlines the project's scope, objectives, deliverables, timelines, resources, risks, communication plan, and quality assurance processes. The Contractor shall utilize project management tools, approved by the Government, to track project progress, manage tasks, and report on key performance indicators, such as schedule adherence, and deliverable quality, and provide updates periodically, or as requested by the Government, to reflect any changes in project scope or approach. The Contractor shall maintain open and transparent communication with the Government throughout the project lifecycle, promptly responding to

inquiries and proactively escalating any potential issues that may arise. All communication shall be documented and accessible to both the Contractor and the Government.

The Contractor will also be responsible for the Program Management activities, including overseeing the overall program strategy, goals, and benefits realization. The Contractor's Program Manager will develop and maintain a comprehensive Program Management Plan that aligns with the Client's strategic objectives and ensures that the projects within the program are delivered on time, within budget, and to the required quality standards. The Program Manager will be responsible for identifying, assessing, and mitigating program-level risks. The Contractor's Program Manager will also facilitate communication and collaboration among stakeholders, including project teams, subcontractors, and the Client, to ensure that program goals and objectives are met. This will include providing regular program status updates, identifying and addressing any program-level issues or concerns, and ensuring that lessons learned are captured and applied across the program. The Contractor's Program Manager will work closely with the Client's designated representative to ensure that the program is delivered in a manner that meets the Client's needs and expectations, and that all program deliverables meet the required quality standards.

Deliverables: Vendor to propose Project Management deliverables including but not limited to monthly status reports.

6.2 - Code Modernization

Code modernization involves transforming legacy application code from outdated programming languages, such as COBOL, Assembler, or PL/I, into modern, maintainable languages like Java, C#, or Python. This process typically includes analyzing existing code to identify bottlenecks, inefficiencies, and outdated constructs, then refactoring or rewriting the codebase to align with modern programming paradigms. These efforts ensure that legacy applications can integrate with modern platforms, cloud environments, and technologies like microservices, containers, and serverless architectures. Code modernization also incorporates best practices in security, scalability, and automation to enhance overall application performance and maintainability.

Approaches to code modernization include automated refactoring, rewriting, and incremental modernization. Automated refactoring uses tools to convert legacy code while preserving functionality, while rewriting involves reengineering applications to utilize modern languages, architectures, and frameworks. Incremental modernization enables phased updates to ensure continuity of operations during transformation.

Deliverables: Vendor to propose Code Modernization deliverables including but not limited to: Modernization Roadmap, Refactored/Converted Codebase, API Integration Documentation, Automated Testing Frameworks, Compliance and Security Reports, Training Materials, and Performance Metrics and Monitoring Dashboards.

6.3 - Code Optimization

Code Optimization is a foundational activity that transforms inefficient, resource-intensive, and brittle code into performant, maintainable, and scalable application components within hybrid and cloud-integrated environments. This task area directly supports the broader goals of modernization by improving application efficiency, reducing technical debt, enhancing maintainability, and enabling the modularization of legacy systems. These efforts align with DISA's mission to deliver operational agility, cost-efficiency, performance reliability, and system sustainability to its Mission Partners.

Optimization efforts begin with deep analysis of legacy and modernized codebases using static and dynamic techniques, including complexity modeling, control and data flow analysis, and runtime diagnostics. Advanced profiling tools are used to observe application behavior under real execution conditions, identifying performance hotspots and inefficient execution paths. Code profiling is combined with scale testing techniques to simulate workloads at increasing magnitudes (such as 1x, 5x, and 10x baselines) to evaluate how performance scales and to quantify the impact of optimization over time. This methodology not only validates code improvements but also provides critical insights into system bottlenecks and opportunities for reengineering.

Large language model-enabled tools play a significant role in accelerating the code comprehension and optimization process. These tools can assist in identifying and documenting hard-coded business rules, redundant logic, and opportunities for code reuse across legacy components. They are also capable of generating semantic maps, developer documentation, and detailed business logic catalogs that provide transparency into systems with minimal

existing documentation. The outputs of these AI-assisted activities are validated by human subject matter experts and contribute directly to sustainment, training, and audit readiness.

A significant focus of code optimization is the structured reduction of technical debt. By applying semi-objective measurement frameworks, the maintainability and complexity of the codebase can be assessed using metrics such as cyclomatic complexity, code duplication, and nesting depth. These insights help prioritize refactoring efforts that yield the greatest benefit to performance and sustainability. Optimization work includes identifying and eliminating dead code, orphaned copybooks, and unreachable logic branches. It also includes aligning code coverage analysis with test plans to ensure that modernization efforts are verifiable, maintainable, and resilient.

Optimization includes targeted refactoring of code to reduce complexity and increase modularity. In many cases, this includes functional decomposition of large monolithic programs into discrete components. Where appropriate, this task area supports the transition to microservices architectures by identifying logical service boundaries, decoupling dependencies, and preparing code modules for standalone deployment. Refactored components are updated to reflect modern coding standards and best practices, ensuring consistency and interoperability across platforms.

All optimization work is performed in coordination with related task areas including Testing and QA, Security and Compliance, Infrastructure Modernization, and Data Modernization. Optimization outputs are fully documented, traceable, and aligned with regulatory and cybersecurity frameworks to ensure that performance gains do not compromise mission assurance.

Deliverables: Vendor to propose Code Optimization deliverables including but not limited to: Hard-Coded Business Rule Discovery and Documentation, User and Reference Guide Generation, Validation, and Update Packages, Developer and Analyst Documentation Packages, Code Optimization Roadmap, Code Profiling and Runtime Behavior Reports, Scale Testing Summary, Code Refactoring Logs, Functional Decomposition and Modularization Plans, Microservices Enablement Roadmap, Code Coverage and Test Effectiveness Reports, Dead Code Elimination Summary, Optimized Codebase, AI-Assisted Refactoring Logs, Business Rule Validation Reports, Data Dictionary and System Reference Updates, and Code Quality Dashboards.

6.4 - Data Modernization

Data modernization activities focus on transforming and optimizing data systems to meet the evolving needs of mission-critical operations and enable digital transformation. These activities modernize how data is stored, accessed, shared, and governed. They ensure scalability, real-time processing, and integration with advanced analytics and AI capabilities. Core activities include migrating legacy databases to modern platforms, integrating structured and unstructured data into centralized data lakes, and implementing cloud-native data architectures that support real-time analytics.

Data modernization support involves comprehensive assessments of existing data environments to identify legacy structures and integration gaps. Phased modernization roadmaps are developed to align with mission-critical objectives, ensuring data migration and governance processes are seamless and compliant. This enables organizations to break down data silos, enhance operational intelligence, and meet evolving DoD directives and federal regulations.

Data migration focuses on transitioning legacy systems, such as hierarchical or network databases, to modern relational or NoSQL platforms, improving accessibility and scalability. Data lake architectures are designed to centralize structured and unstructured data, providing a flexible and scalable environment for analysis and integration. Robust frameworks for data governance and security ensure compliance, privacy, and access control, adhering to stringent DoD and federal standards. Real-time data enablement transforms data systems to support AI-driven analytics and operational decision-making.

Deliverables: Vendor to propose Data Modernization deliverables including but not limited to: Data Modernization Roadmap, Data Lake Implementation, Data Governance Framework, Real-Time Data Pipelines, Data Analytics Dashboards, Data Migration Toolkits, AI-Enabled Data Solutions, and Compliance and Security Reports.

6.5 - Database Upgrade

Database upgrade activities focus on modernizing legacy databases to address scalability, flexibility, and performance challenges while integrating with modern technologies like cloud computing and advanced analytics platforms. Many Strategic Partners rely on legacy databases such as IDMS, SUPRA, or IMS, which, while historically robust, face limitations in handling real-time data processing and supporting modern applications. Database upgrades aim to overcome these constraints through activities like enhancing legacy database capabilities, migrating to modern relational or NoSQL platforms, and employing purpose-built database technologies tailored to specific workloads.

Database modernization support includes evaluating existing database environments, developing tailored migration strategies, and implementing solutions that align with Strategic Partners' operational goals. Modernized databases improve data accessibility, enable real-time analytics, and reduce operational costs associated with legacy infrastructure. A robust focus on compliance ensures adherence to DoD standards and federal regulations throughout the modernization process.

Database upgrades typically involve upgrading legacy databases to newer versions, migrating to modern platforms, and implementing policy-driven data archiving strategies to optimize storage costs while maintaining compliance with data retention mandates. Additionally, Strategic Partners benefit from purpose-built database solutions, including relational databases for structured data, NoSQL databases for unstructured and large-scale data, graph databases for relationship analysis, and key-value databases for rapid data retrieval.

Deliverables: Vendor to propose Database Upgrade deliverables including but not limited to: Database Modernization Strategy, Modernized Database Platforms, Data Warehousing and Analytics Solutions, Policy-Driven Data Archiving Frameworks, Security and Compliance Documentation, Post-Migration Support Plans, and Integration Documentation.

6.6 - API Enablement

API enablement activities focus on modernizing legacy systems by exposing their functionality and data through Application Programming Interfaces (APIs). This approach enables legacy applications and data to interact with modern systems, applications, and cloud services without the need to fully rewrite or replace legacy code. APIs serve as bridges between legacy systems and modern environments, facilitating real-time data access, decision-making, and integration with emerging technologies such as AI and machine learning.

API enablement support includes evaluating legacy systems for API exposure, designing secure and scalable APIs, and implementing hybrid data ecosystems that integrate legacy applications and cloud environments. These initiatives align with the Pace Layering model, allowing for incremental modernization while preserving mission-critical functionalities.

API enablement enhances scalability, flexibility, and security by integrating mainframe or legacy systems with modern platforms. It democratizes data access, enabling Strategic Partners to leverage advanced analytics and AI tools. Furthermore, this phased modernization approach minimizes risks associated with full system overhauls.

Deliverables: Vendor to propose API Enablement deliverables including but not limited to: API Enablement Roadmap, API Gateway Deployment, API Documentation and Developer Portals, Cloud Integration and Hybrid API Management, Security and Compliance Reports, and API Performance Dashboards.

6.7 - Infrastructure Modernization

Infrastructure modernization activities focus on transforming legacy mainframe and IT infrastructures into dynamic, cloud-aligned environments that integrate with hybrid cloud platforms and automation technologies. By implementing Infrastructure-as-Code (IaC) solutions, re-platforming legacy systems, and aligning with NIST cloud computing guidelines, these activities ensure that mainframes and legacy systems maintain their inherent strengths while adopting modern capabilities.

Infrastructure modernization support includes comprehensive infrastructure assessments, IaC implementations for automated resource provisioning, and the development of hybrid architectures that seamlessly bridge on-premises

and cloud resources. These efforts enhance operational agility, cost efficiency, and scalability, enabling Strategic Partners to dynamically respond to evolving workload demands and technology trends.

Key enhancements in infrastructure modernization include re-platforming applications onto hybrid architectures, deploying automated resource management solutions, and enhancing mainframes and legacy systems to meet NIST cloud computing characteristics, such as rapid elasticity and metered services.

Deliverables: Vendor to propose Infrastructure Modernization deliverables including but not limited to: Infrastructure Modernization Roadmap, NIST Cloud Compliance Assessment, Hybrid Infrastructure Integration, Performance and Security Reports, and Monitoring Dashboards.

6.8 - Performance Optimization

Performance Optimization activities focus on enhancing the efficiency, scalability, and resource utilization of IT systems such as IBM Z, ensuring they can meet the increasing demands of mission-critical workloads. These efforts include fine-tuning infrastructure, applications, and databases to eliminate bottlenecks, improve response times, and maintain seamless operations. By leveraging advanced resource management and hybrid cloud integration capabilities, performance optimization ensures optimal service delivery while controlling costs.

Performance Optimization support encompasses strategies such as workload management, database optimization, and code refactoring. Infrastructure tuning maximizes CPU and memory efficiency, while database optimization enhances query performance and enables faster data retrieval. Code-level adjustments, such as optimizing COBOL and Assembler logic, reduce processing times and improve application responsiveness. Hybrid cloud offloading allows non-critical tasks to be dynamically allocated to scalable cloud resources, ensuring systems prioritize essential operations.

Additionally, Performance Optimization integrates real-time monitoring tools that provide continuous insights into system performance, enabling proactive adjustments and minimizing downtime. These activities not only improve operational efficiency but also ensure alignment with compliance standards and future scalability.

Deliverables: Vendor to propose Performance Optimization deliverables including but not limited to: Performance Optimization Roadmap, Optimized DB2 Database Environment, Code Optimization and Refactoring, Hybrid Cloud Integration Plan, Real-Time Performance Monitoring Tools, and Optimization and Security Reports.

6.9 - Hybrid Cloud Integration

Hybrid Cloud Integration activities focus on synchronizing on-premises mainframe and on-prem systems with cloud platforms, enabling scalable, secure, and flexible operations. This ensures mission-critical workloads remain secure on the right-fit platform while leveraging the agility of cloud-native solutions for analytics, AI, and other compute-intensive tasks. By integrating hybrid architectures, these activities enhance real-time data sharing, operational efficiency, and compliance with DoD standards.

Hybrid Cloud Integration support includes designing architectures that bridge on-prem and cloud environments, enabling seamless data exchange and workload distribution. These efforts reduce operational silos, improve scalability, and optimize cost efficiency through the adoption of cloud-native technologies. Additionally, the integration of hybrid cloud platforms supports Strategic Partners' needs for secure, dynamic, and future-ready IT ecosystems.

Deliverables: Vendor to propose Hybrid Cloud Integration deliverables including but not limited to: Hybrid Cloud Integration Roadmap, Real-Time Data Integration Frameworks, API Integration and Management, Hybrid Cloud Monitoring Dashboards, and Security and Compliance Reports.

6.10 - Testing and Quality Assurance (QA)

Testing and Quality Assurance (QA) activities ensure that modernized systems, applications, and infrastructure meet functional, performance, and security requirements, supporting mission-critical operations. These activities are embedded throughout the modernization lifecycle, from the initial refactoring of legacy code to the final stages of User Acceptance Testing (UAT). Comprehensive testing strategies validate that all components work seamlessly together, mitigate risks, and enhance system reliability. Leveraging frameworks like Test-Driven Development

(TDD), Behavior-Driven Development (BDD), and automation tools such as Selenium, QA ensures thorough validation with increased efficiency. Additionally, generative AI accelerates the process by automating test case generation and identifying edge cases, reducing manual effort.

Testing and Quality Assurance support includes developing tailored testing strategies aligned with modernization goals, implementing automated frameworks for continuous testing, and providing performance and security validation to ensure systems meet stringent compliance standards. This approach integrates with DevOps practices through Continuous Integration and Continuous Delivery (CI/CD), enabling rapid feedback cycles and minimizing risks during deployments.

Deliverables: Vendor to propose Testing and Quality Assurance (QA) deliverables including but not limited to: Comprehensive Testing Strategy Document, Generative AI-Driven Test Case Library, Automated Test Suites, Performance Testing Reports, UAT Scripts and Results, Security and Compliance Reports, and Real-Time Test Monitoring Dashboards.

6.11 - Security and Compliance

Security and Compliance activities are foundational to any modernization effort, particularly in environments supporting mission-critical and sensitive workloads. These activities ensure that systems, applications, and data comply with DoD, DISA, and federal regulations while upholding the highest cybersecurity standards. Security and compliance support spans integrating advanced Identity, Credential, and Access Management (ICAM) frameworks, achieving Authority to Operate (ATO), and addressing challenges such as the System and Network Authorization Process (SNAP) and Cloud Permission to Connect (CPTC) approvals.

Security modernization involves extending traditional security tools like RACF (Resource Access Control Facility) into modern Role-Based Access Control (RBAC) systems for hybrid architectures. This integration ensures seamless access control, authentication, and authorization across platforms. ICAM solutions further support transitioning to Zero Trust (ZT) architectures, enhancing continuous identity verification and secure access across hybrid environments.

Through streamlined processes for ATO and CPTC, Security and Compliance activities simplify compliance tracking using tools like SNAP and eMASS. These efforts reduce the risks of unauthorized access while ensuring consistent security governance for both on-premises and cloud-native systems. This approach enables Strategic Partners to confidently modernize their systems while adhering to stringent DoD and federal security requirements.

Deliverables: Vendor to propose Security and Compliance deliverables including but not limited to: Security and Compliance Assessment Reports, ICAM Implementation Frameworks, Zero Trust Architecture Deployment Plans, ATO and CPTC Roadmaps, Security Incident Response Plans, Penetration Testing and Vulnerability Assessment Reports, RBAC Integration Documentation, and Compliance Certification Reports.

6.12 - Training and Documentation

Training and Documentation activities focus on equipping Strategic Partners with the skills, knowledge, and tools necessary to manage and maintain modernized systems effectively. These activities ensure seamless adoption of new technologies and processes by leveraging advanced instructional design methodologies and generative AI for content development. Training programs address skill gaps caused by modernization, while comprehensive documentation ensures continuity, troubleshooting, and optimization of modernized systems.

Training support includes developing tailored learning modules based on frameworks such as ADDIE and Bloom's Taxonomy to address the varying skill levels of personnel. Scenario-based training enables hands-on practice in simulated environments, bridging the gap between theoretical knowledge and real-world application. Generative AI accelerates the creation and maintenance of both training materials and system documentation, dynamically updating content as systems evolve. This ensures accuracy and reduces onboarding time for new personnel.

Documentation efforts focus on creating detailed, user-friendly manuals, system guides, and code-level documentation that are dynamically maintained with changes in system architecture. These resources ensure Strategic Partners can independently manage their systems while maintaining compliance with operational standards.

Deliverables: Vendor to propose Training and Documentation deliverables including but not limited to: Tailored Training Program Roadmap, Generative AI-Produced Documentation, Interactive Training Modules, Training Maturity Model Evaluation Reports, and Training Performance Dashboards.

6.13 - Cutover Planning and Execution

Cutover Planning and Execution activities focus on the seamless transition of legacy systems, applications, or infrastructure to modernized environments. These processes are meticulously designed to ensure minimal disruption, preserve data integrity, and maintain operational continuity. A structured cutover plan addresses all phases, including pre-cutover, execution, and post-cutover activities, to manage risks and validate the success of the migration.

Cutover Planning and Execution support includes developing comprehensive plans that detail each step of the migration process, including rehearsals, stakeholder coordination, and backout strategies. Pre-cutover activities involve detailed testing and simulations to validate the cutover approach, ensuring readiness before initiating the transition. Post-cutover hypercare provides continuous monitoring, rapid resolution of issues, and validation of system stability, ensuring a smooth handoff from the transition team to operational control. Legacy systems are decommissioned as part of the final phase to ensure all resources are reallocated effectively.

These activities mitigate risks during large-scale migrations, such as those involving hybrid cloud architectures, and provide Strategic Partners with the tools and guidance necessary to achieve successful transitions with minimal downtime or data loss.

Deliverables: Vendor to propose Cutover Planning and Execution deliverables including but not limited to: Detailed Cutover Plan and Go/No-Go Criteria, Rehearsal and Simulation Reports, Hypercare Support Plan, and Decommissioning and Legacy Cleanup Plan.

6.14 - Data Architecture Support

Data Architecture Support establishes the foundation for managing data as a critical asset across mission-critical systems. Modern data architecture is designed to be scalable, adaptable, and responsive to evolving operational needs, ensuring compliance with DoD Data Strategy principles. A key focus is creating a nimble architecture that supports DataOps and real-time data pipeline orchestration, allowing rapid integration of new data sources, enhanced interoperability, and advanced analytics capabilities.

Data Architecture activities replace outdated, point-to-point integration patterns with modern, metadata-driven configurations that simplify data exchanges and eliminate unnecessary complexity. By adopting strict decoupling between compute and storage, this approach enhances system flexibility, enabling seamless integration with hybrid and cloud environments. Such flexibility supports advanced AI/ML use cases and real-time decision-making, optimizing the utility of data assets across diverse operational scenarios.

Data Architecture Support also emphasizes the importance of strong data governance as a guiding principle, ensuring all changes to the architecture adhere to mission-critical goals and compliance standards. Strategic Partners benefit from architectures designed to be agile, scalable, and efficient while maintaining high data quality and security standards.

This approach equips Strategic Partners with a data architecture designed to evolve alongside technological advancements, supporting operational agility, and maximizing the value of data assets.

Deliverables: Vendor to propose Data Architecture Support deliverables including but not limited to: Data Architecture Roadmap (DARM), DataOps Implementation Plan, API Service Registry and Data Catalog, and AI-Assisted Metadata Management Plan.

6.15 - Solution Architecture Support

Solution Architecture Support provides Strategic Partners with expert guidance to design, integrate, and optimize systems across diverse platforms and domains, ensuring alignment with strategic goals and operational needs. This service encompasses cross-platform integration, domain-specific support, and the optimization of existing

environments, such as mainframe, cloud, analytics, or API ecosystems. By managing technical complexities, solution architecture ensures scalability, efficiency, and seamless interoperability while reducing technical debt.

Solution architects leverage frameworks like Enterprise Architecture (EA) to align business process, data, application, and infrastructure layers. This alignment ensures systems are designed to support organizational workflows and collaboration models effectively. Solution architecture also reduces long-term costs associated with ad hoc fixes by addressing technical debt and applying industry best practices. Cross-platform integration, such as connecting mainframes to cloud environments, highlights the versatility of solution architecture. Conversely, domain-specific expertise ensures optimal performance and modernization within focused areas, such as enhancing legacy mainframes to integrate with modern analytics or API tools.

Deliverables: Vendor to propose Solution Architecture Support deliverables including but not limited to: Mainframe-Specific Architecture Blueprint, Technical Debt Assessment and Refactoring Plan, and Enterprise Architecture Artifacts.

6.16 - Data Governance and Policy Support

Data Governance and Policy Support establishes a structured framework for managing data as a critical organizational asset. By defining policies, standards, and procedures, this task area ensures that data remains accessible, secure, and aligned with mission-critical objectives. Governance provides the foundational "golden rules" that guide how data is stored, accessed, and utilized, ensuring compliance with regulatory standards and alignment with the DoD Data Strategy and VAULTIS principles. This governance-driven approach also ensures that evolving data architectures remain adaptable, scalable, and interoperable across hybrid environments.

Data Governance activities focus on enhancing data quality, visibility, and lifecycle management while supporting the integration of advanced analytics and AI capabilities. Governance frameworks ensure that architectural changes, such as migrations or modernizations, are guided by clear standards to maintain data integrity and mission readiness. These efforts enable Strategic Partners to manage data proactively, ensuring it remains a strategic asset capable of supporting real-time decision-making and operational excellence.

Deliverables: Vendor to propose Data Governance and Policy Support deliverables including but not limited to: Data Governance Framework, Governance-Driven Data Architecture Roadmap, Policy Compliance Analysis, Data Quality and Metadata Management Plan, and Training Materials for Governance Standards.

6.17 - Vendor and Program Support

Vendor and Program Support involves managing and overseeing third-party providers engaged in modernization efforts, ensuring their services align with mission-critical objectives and meet required standards. This task area emphasizes the role of a trusted broker to help Strategic Partners identify, evaluate, and select reliable vendors while providing independent oversight to ensure compliance with governance, performance, and security requirements throughout the project lifecycle.

Vendor and Program Support includes structured program management that integrates Governance, Risk, and Compliance (GRC) models to mitigate risks and ensure efficient resource allocation. Agile delivery methodologies provide flexibility, enabling modernization efforts to adapt to evolving mission needs without compromising strategic objectives. Through these activities, Strategic Partners benefit from transparent oversight, streamlined vendor operations, and alignment with modernization goals.

Deliverables: Vendor to propose Vendor and Program Support deliverables including but not limited to: Vendor Selection Reports, Independent Oversight Plans, Program Management Frameworks, and Agile Delivery Playbooks.

6.18 - Hybrid Cloud Integration and Optimization

Hybrid Cloud Integration and Optimization activities aim to create a seamless, secure, and scalable infrastructure that combines the robustness of on-premises IT and mainframe systems with the flexibility of cloud platforms. This strategy enables Strategic Partners to optimize workload distribution by retaining critical operations on secure on-prem systems while offloading less critical, compute-intensive tasks to public or private clouds. The approach ensures operational efficiency, enhanced scalability, and cost optimization through dynamic resource allocation.

The integration strategy includes leveraging the Pace Layering model, allowing mission-critical systems of record to remain stable on systems like mainframes, while enabling faster-moving components to evolve on cloud platforms. This model supports seamless API enablement, ensuring real-time interaction between on-premises systems and cloud-based applications. Additionally, hybrid cloud solutions enhance resilience by integrating cloud-based disaster recovery options, ensuring continuity during unexpected events.

These activities provide Strategic Partners with a future-ready infrastructure capable of supporting real-time analytics, AI/ML workloads, and edge computing, ensuring flexibility and scalability to meet evolving operational demands.

Deliverables: Vendor to propose Hybrid Cloud Integration and Optimization deliverables including but not limited to: Hybrid Cloud Integration Roadmap, Workload Offloading and Optimization Plan, API Integration Framework, Disaster Recovery and Resilience Strategy, and Performance Monitoring Dashboards.

6.19 - Mainframe AI and Automation Integration

Mainframe AI and Automation Integration activities enhance operational efficiency, predictability, and decision-making capabilities by embedding artificial intelligence (AI) and automation into mainframe workflows. These efforts allow mission-critical applications to operate seamlessly, leveraging technologies such as IBM z16's on-chip AI acceleration. AI-driven automation supports routine task management, such as resource allocation and workload optimization, enabling mainframe systems to operate more predictably while minimizing the need for manual intervention.

This approach ensures that AI and machine learning (ML) models are optimized for real-time decision-making, anomaly detection, and predictive analytics directly within mainframe environments. The integration of AIOps (Artificial Intelligence for IT Operations) enables Strategic Partners to proactively address system issues, enhance system resiliency, and improve uptime. These activities align with the overarching goal of modernizing mainframe systems while maintaining high levels of security and compliance.

Deliverables: Vendor to propose Mainframe AI and Automation Integration deliverables including but not limited to: AI-Driven Automation Framework, Real-Time AI Model Deployment, AIOps Implementation Plan, Performance Monitoring Dashboards, and AI Model Optimization Reports.

6.20 - Sustainability-Driven Legacy Modernization

Sustainability-Driven Legacy Modernization focuses on transforming legacy systems to achieve significant energy efficiency, reduce environmental impact, and optimize physical data center space. These activities include migrating workloads to energy-efficient mainframe platforms, implementing carbon footprint monitoring, and optimizing resource allocation to align with organizational sustainability goals. By utilizing modern mainframes such as IBM Z or LinuxONE, organizations can achieve substantial energy savings while maintaining superior performance.

This approach involves consolidating workloads onto fewer, more efficient systems, reducing physical space requirements and infrastructure costs, such as cooling and power consumption. In addition, real-time monitoring tools enable Strategic Partners to track energy consumption and carbon footprint, providing transparency and supporting compliance with environmental regulations. By modernizing mainframe infrastructures with a focus on sustainability, organizations can reduce operational costs and align with broader environmental goals.

Deliverables: Vendor to propose Sustainability-Driven Legacy Modernization deliverables including but not limited to: Energy Efficiency Assessment Report, Workload Consolidation Plan, Sustainability Metrics Dashboard, and Compliance and Sustainability Reports.

6.21 - Mainframe Workforce Development

Mainframe Workforce Development focuses on bridging the skills gap within the mainframe workforce by equipping personnel with the expertise required to manage and maintain modernized systems. This involves targeted training, mentorship programs, and leveraging advanced tools to enable the workforce to effectively handle AI-driven technologies, cloud-native platforms, and automation frameworks. The initiative ensures Strategic Partners have a skilled, future-ready workforce capable of sustaining operational efficiency in hybrid and advanced technological environments.

By addressing the declining availability of legacy system expertise, this support enables organizations to transition from dependency on outdated skills to cultivating a workforce proficient in modern technologies. Key efforts include hands-on training, scenario-based simulations, and mentorship opportunities to ensure knowledge transfer. Additionally, leveraging AI tools for personalized learning accelerates workforce readiness, ensuring rapid adaptation to modernized systems.

Deliverables: Vendor to propose Mainframe Workforce Development deliverables including but not limited to: Upskilling Training Program Roadmap, Interactive Training Modules, Mentorship and Knowledge Transfer Plans, AI-Enhanced Learning Tools, and Skills Gap Analysis Reports.

6.22 - Mainframe-Driven Analytics and AI Enablement

Mainframe-Driven Analytics and AI Enablement activities empower Strategic Partners to leverage advanced analytics, artificial intelligence (AI), and machine learning (ML) capabilities directly within mainframe environments. By integrating AI models and analytics workflows into mainframes, organizations can process vast amounts of data in real time, gaining actionable insights while maintaining the robust security and reliability of mainframe systems.

These efforts include deploying AI and analytics tools optimized for mainframe workloads, enabling predictive maintenance, anomaly detection, and operational decision-making. The integration of hybrid analytics platforms further supports cross-platform data analysis, ensuring seamless collaboration between mainframe and cloud environments. These activities align with the DoD's focus on data-centric decision-making and operational excellence.

Deliverables: Vendor to propose Mainframe-Driven Analytics and AI Enablement deliverables including but not limited to: Mainframe AI Enablement Plan, Hybrid Analytics Integration Framework, AI Model Deployment Guidelines, Analytics Dashboard Suite, and Compliance and Security Reports for AI Workloads.

6.23. Other Direct Costs (ODCs)

The Contractor may be required to purchase incidental hardware or software tools, supplies, or materials to complete projects. ODCs will be proposed for each task or call order.

7. Performance Standards. Vendor to propose Performance Standards for each of the tasks identified in section 6.

8. Place of Performance. The place of performance for services will be agreed upon by all parties and must be pre-approved in writing by the COR. With COR approval, work may be performed remotely, at a DISA location, or at location provided by a strategic partner. Secret or TS/SCI work will have to be done at a DISA location. Remote work cannot be done on Top Secret/SCI workloads.

Mission Location: DISA, Mechanicsburg, PA

9. Period of Performance. The contract will be a five-year ordering period. Task or call orders will have an individual period of performance based on the project scope and schedule.